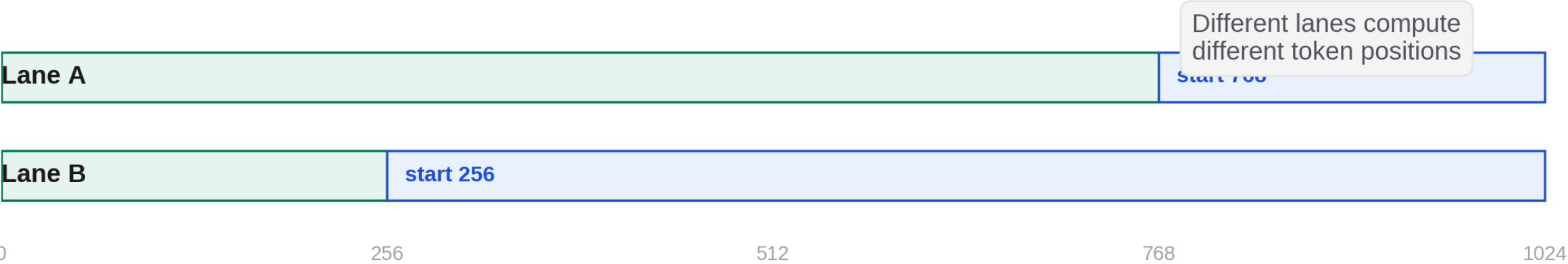


Common-start is correct—but recomputes longer prefix hits

Current: one common start for the whole batch

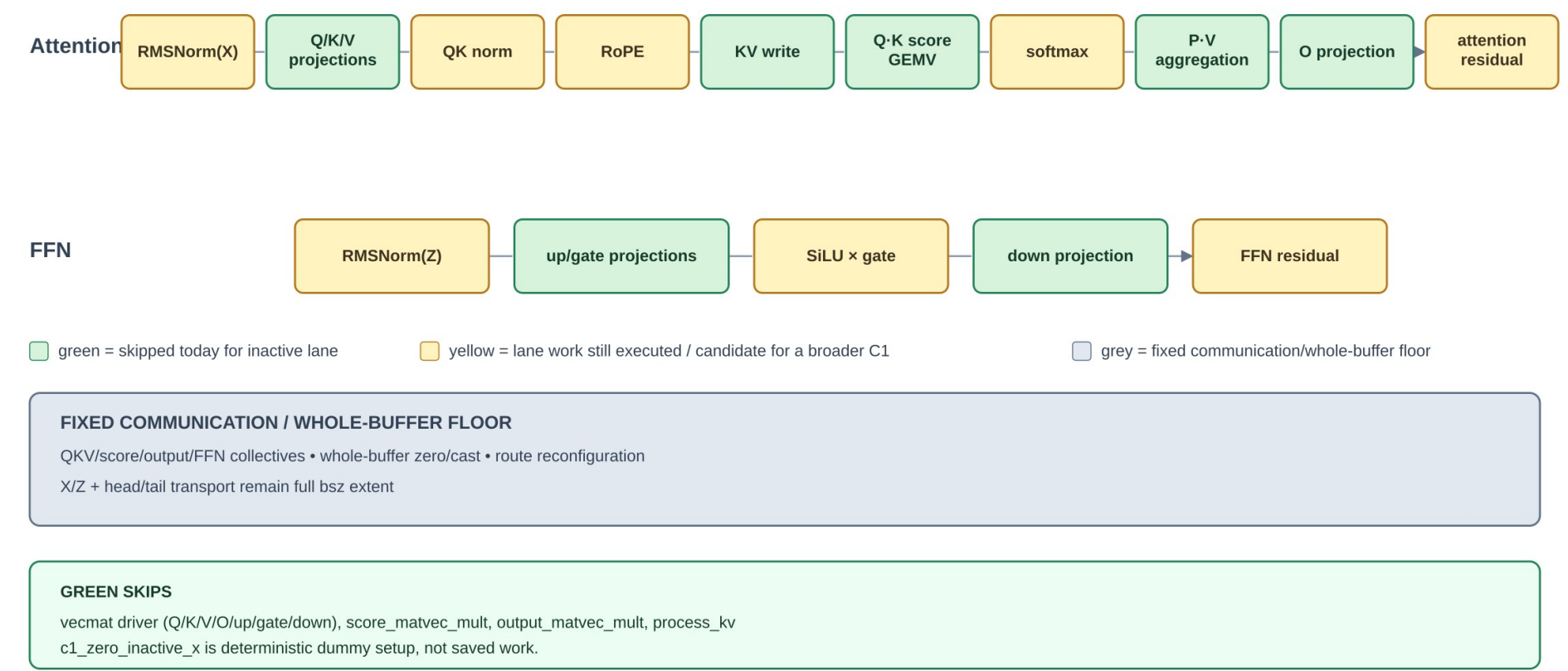


Needed: per-lane cursors inside the same round



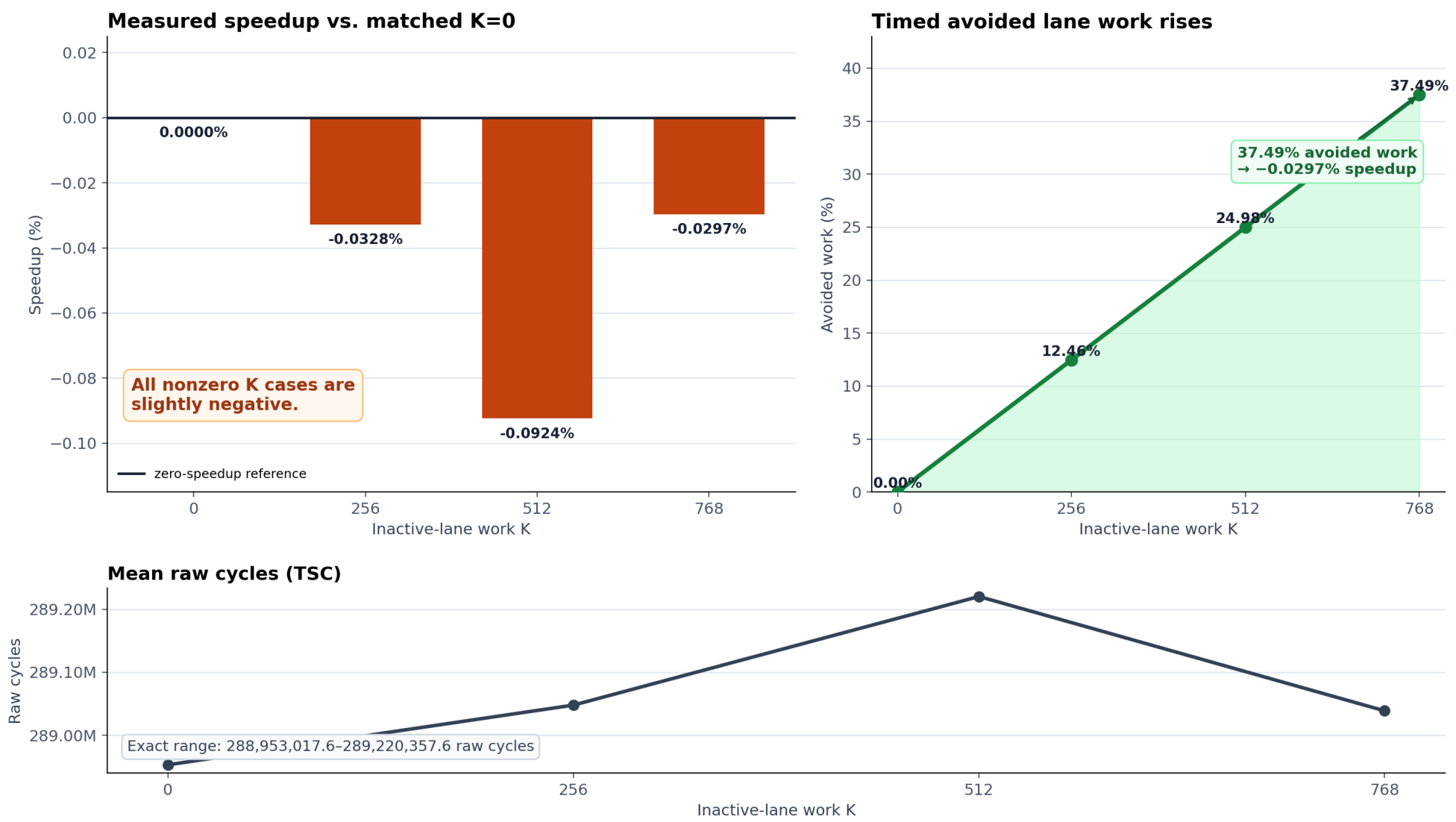
Lane A can reuse 768 tokens while Lane B can reuse only 256. The current batch starts both at 256, so Lane A re-executes 512 already-known token positions.

We skipped lane-sliced compute; the fixed batch floor stayed



Green is predicated while $\text{step} < R_i - \min(R)$; yellow still executes for the inactive lane; grey remains full-bsz communication or whole-buffer work.

37.49% less lane-step work produced no critical-path speedup



Full Qwen3-1.7B, bsz=2, S/N/F/G=256/1024/769/255, five repetitions per case, raw TSC cycles. At K=768, mean speedup is -0.0297%.

Pin the fixed floor—or move to independent lane progress

A · Profile the C1 fixed floor

- Time or ablate collectives, whole-buffer ops and transport by phase
- Identify the single largest remaining critical-path item
- Broaden masking only if that removable term can change the $<0.1\%$ result

B · Design C2 / continuous batching

- Independent per-lane start, cursor and completion state
- Price added SRAM, protocol, admission and scheduler complexity
- Only credit gains that survive a serving-like schedule replay